# Chapter 3 Notes

## 3.1

In the proof of Theorem 3.1, here is why the class of all transitive natural numbers is a set. We can show that it is a subset of $\omega$, and is hence a set by $A_2$. For suppose $x$ is a member of the class of all transitive natural numbers. Then obviously $x$ is also a natural number, hence $x \in \omega$, thus completing the proof.

Note that here is why the only natural numbers allowed for by Definition 2.2 are the ones defined right before Definition 1.1. By Definition 2.2, a natural number is a member of every inductive set. Now, consider the following inductive set: $\emptyset$ is an element, and if $x$ is an element, then so is $x^+$. Hence every natural number is a member of this set. But this set is just the set of all sets that were defined in the way just before Definition 1.1.

### Exercise 3.1

Suppose that $A$ is a class of natural numbers such that for every natural number $n$, if $A$ contains all numbers less than $n$, then $A$ contains $n$ as well. Now, we can prove by induction that $A$ contains all natural numbers. Consider the property $P$ meaning $A$ contains all numbers natural numbers less than or equal to $n$. Now, there are no natural numbers less than 0. Thus the statement that $A$ contains all natural numbers less than 0 is vacuously true; hence $A$ contains 0. Thus, because 0 is the only natural number less than or equal to 0, $P$ holds for 0. Now suppose that $P$ holds for some $n$, that is, $A$ contains all natural numbers less than or equal to $n$. But then $A$ contains all numbers less than $n+1$, meaning (by the definition of $A$) $A$ also contains $n+1$. Thus, by induction, $A$ contains all natural numbers.

### Exercise 3.2

Consider a natural number $n$.

(a): Suppose that $x \in n$. Now, it cannot be that $n = \emptyset$. Thus $n$ is non-empty, hence of the form $(n-1)^+$. Thus either $x \in n-1$ or $x = n-1$. Now, $\cup n$ is just the set of all elements of all elements of $n-1$ together with all elements of $n-1$. Now suppose $x = n-1$. Then $x \notin \cup n$. For suppose it were. Then it would either be an element of $n-1$ or it would be an element of an element of $n-1$. In the first case, we would have $n-1 \in n-1$, which is ruled out by Theorem 3.2. In the second case, there would be some $y$ such that $x \in y$ and $y \in n-1$. But then, by the transitivity of $n-1$ (by Theorem 3.1), $x \in n-1$, that is, $n-1 \in n-1$, which is again ruled out by Theorem 3.2. Thus $n-1 \notin \cup n$, meaning the statement is not necessarily true.

It should be noted that all elements other than $n-1$ of $n$ are members of $\cup n$. For all elements of $n$

are either elements of $n-1$ or equal to $n-1$. Thus, if they are not equal to $n-1$, the element is an element of $n-1$. But, because $n-1 \in n$, then all elements of $n-1$ are elements of $\cup n$.

(b): Suppose that $x \in \cup n$. Thus either $x \in n-1$ or $x$ is an element of some element of $n-1$. In the first case, because $n-1 \in n$, $x \in n$ (by the transitivity of $n$). In the second case, because $n-1$ is transitive, it follows that $x \in n-1$. But then, because $n-1 \in n$ and $n$ is transitive, $x \in n$. Thus, in either case, $x \in n$, and thus $\cup n \subseteq n$. Hence the statement is necessarily true.

(c): By part (b) of this problem, it follows that $x \in \cup n$ implies $x \in n$. But $n^+ = n \cup \{n\}$. Thus all elements of $n$ are in $n^+$, meaning $x \in n^+$. On the other hand, suppose that $x \in n^+$. Then it could be that $x = n$. But then, if $x \in \cup n$, we would have either $n$ being an element of an element of $n-1$, or $n \in n-1$. In the first case, the transitivity of $n-1$ would show that $n \in n-1$. Thus we would have, in either case, $n \in n-1$, which is impossible. For if $n \in n-1$, then, because $n-1 \in n$, $n-1 \in n-1$ by the transitivity of $n-1$, which is impossible (or, if $n-1 = \emptyset$, it could have no members). Hence it is not necessarily the case that $n^+ \subseteq \cup n$. Thus $\cup n = n^+$ does not necessarily hold.

Similar reasoning as to the above also shows that $n-1$ cannot be an element of $\cup n$. It should be noted that $n-1$ and $n$ are the only members of $n^+$ that are not members of $\cup n$. For any other element of $n^+$ is an element of $n$ but not equal to $n-1$. But $n = n-1 \cup \{n-1\}$. Hence it would be an element of $n-1$, and because $n-1$ is an element of $n$, it would also be an element of $\cup n$.

(d): On the one hand, suppose that $x \in \cup n^+$. Then either $x$ is an element of an element of $n$, or $x$ is an element of $n$. In the first case, because $n$ is transitive, $x \in n$. The second case means $x \in n$. Thus, in any case, $x \in n$. On the other hand, suppose that $x \in n$. Then $x \in \cup n^+$, as $n^+ = n \cup \{n\}$. Hence $\cup n^+ = n$.

(e): By part (b) of this problem, $\cup n \subseteq n$ holds. But by the answer to part (a) of this problem, it does not necessarily follow that $n \subseteq \cup n$. Thus it is not necessarily true that $\cup n = n$.

## Exercise 3.3

(a): On the one hand, suppose that $x \in \cup\omega$. Then $x$ is an element of a natural number. Thus either $x \in n$ or $x = n$. Now we prove the following statement: every element of every natural number is a natural number. The base case vacuously holds, as $\emptyset$ has no elements. Now suppose the statement holds for some $n$. Then $n^+ = n \cup \{n\}$. Hence, if $y \in n^+$, either $y \in n$ or $y = n$. In the former case, by the induction assumption, it follows that $y$ is a natural number, and in the latter case, $y$ would be a natural number (as $n$ is a natural number). Thus, in any case, $y$ is a natural number, therefore completing the proof. As a consequence of having proven this statement, it follows that $x$ is a natural number, and hence $x \in \omega$. On the other hand, suppose that $x \in \omega$. Thus, for some $n$, $x = n$. Now, all elements of $n^+$ are in $\cup\omega$. But $n$ is an element of $n^+$. Thus $n \in \cup\omega$, and hence $x \in \cup\omega$. Thus we have proven that $\cup\omega = \omega$.

(b): On the one hand, suppose that $x \in \cup\omega^+$. Now, $\omega^+ = \omega \cup \{\omega\}$. Thus either $x$ is an element of a natural number or $x$ is a natural number. By the results of part (a) of this problem, it follows that if $x$ is an element of a natural number, then $x$ is a natural number. Thus $x$ is a natural number, that is, $x \in \omega$. Now suppose that $x \in \omega$. Then $x \in \cup\omega^+$, as $x$ would be an element of $\omega$, which is an element of $\omega^+$. Thus we have proven that $\cup\omega^+ = \omega$.

Note that a generalization cannot be made of this problem. Define $\{\omega_0, \omega_1, \omega_2, ..., \omega_n, ...\}$ in the following way: $\omega_0 = \omega$ and $\omega_{n+1} = \omega_n^+$. Then it does not follow that $\cup\omega_n = \omega$ for all $n$. For consider $n = 2$. This is $\omega_1^+$, that is,

$$\omega_1 \cup \{\omega_1\} = (\omega \cup \{\omega\}) \cup \{(\omega \cup \{\omega\})\}$$

Thus $\omega \in \cup\omega_1^+$. But $\omega \notin \omega$, meaning $\cup\omega_2 \neq \omega$. Indeed, it is a general fact that, for $n \geq 2$, $\cup\omega_n \neq \omega$. To prove this, we can just prove that, for all $n \geq 2$, $\omega \in \omega_n$. We have just proven the base case. If the result

holds for some $n$, then $\omega \in \omega_{n+1}$, as $\omega_{n+1} = \omega_n \cup \{\omega_n\}$, and it was assumed that $\omega \in \omega_n$. Thus, for every $n \geq 2$, $\omega \in \omega_n$. Now we can prove the statement that, for $n \geq 2$, $\cup\omega_n \neq \omega$. We proved the base case for $n = 2$ above. Now consider some $n > 2$. Then $\omega_n = \omega_{n-1} \cup \{\omega_{n-1}\}$ and $\cup\omega_n$ consists of all elements of all elements of $\omega_{n-1}$ and all elements of $\omega_{n-1}$. Because $\omega$ is an element of $\omega_{n-1}$, it follows that $\omega \in \cup\omega_n$, and thus, because $\omega \notin \omega$, $\cup\omega_n \neq \omega$.

# Section 4

## Notes

Note that $g$-induction only works on minimally inductive sets for the following reason: If $A$ was not minimally inductive under $g$, we would have some element $x$ of $A$ such that $x$ is not a term of the $g$-sequence (see the section below titled Exercise 4.1) $\emptyset, g(\emptyset), g(g(\emptyset)), ...$ But the proof by $g$-induction only proves the result for the terms of the $g$-sequence started by $\emptyset$, meaning we would not necessarily know that the result held for $x$.

Note that (1) of Lemma 4.6 holds because 0 is just $\emptyset$, and $\emptyset$ obviously is a subclass of $y$ for every $y$.

## Theorem 4.16

The proof of Theorem 4.16 seems incorrect to me. At a point in the proof, we rely upon Theorem 4.12. But that assumes $L$ is a bounded *subset* of $A$. However he assumes that $\emptyset \notin A$ and shows that $\emptyset \in L$, meaning $L$ is not a subset of $A$, contrary to the claim that it was. (In fact, I think we would need to show that $L$ is a subset of $M$, as $M$ is the one which is minimally inductive under $g$, but Smullyan does not make this clear)

I think that what Smullyan was trying to say was: Let $L$ be the class of all elements of $x$ *of* $M$ such that $x$ is a proper subclass of all elements of $A$. Obviously, this means that $L$ is a subset of $M$, and then, when we show that $L$ is bounded by $A$, we really can use Theorem 4.12. Also, when we assume that $\emptyset \notin A$, we can still conclude that $\emptyset \in L$, as $\emptyset \in M$ by the assumption that $M$ is minimally inductive under $g$.

## Exercise 4.1

We suppose that $N$ is closed under $g$, where $g$ is progressing, and that $b$ is an element of $N$ and that no proper sublcass of $N$ containing $b$ is closed under $g$. In order to prove the desired results, we can first prove (1) of Theorem 4.17. We are given that $N$ is closed under $g$. Call a sequence of the form $x, g(x), g(g(x)), g(g(g(x))), ...$ a $g - sequence$ (and call it a subset of a set $A$ if all elements of the sequence are elements of $A$). Label $x$ the first term of the $g$-sequence, $g(x)$ the second term, and, in general, if $y$ is the $n$th term of the $g$-sequence, then $g(y)$ is the $n + 1$th term of the $g$-sequence (and we say that $n$ is the *index* of the $n$th term of the sequence). We first show that $N$ is just the $g$-sequence $b, g(b), g(g(b)), ...$ Now, it is obvious that the $g$-sequence started by $b$ is a subset of $N$ (because $N$ is closed under $g$, and $b$ is in $N$). Then suppose that there was some element $c \in N$ outside of the $g$-sequence started by $b$. Then the $g$-sequence $c, g(x), g(g(c)), ...$ would be a subset of $A$. But then $b, g(b), g(g(b)), ...$ would be a proper subset of $N$ containing $b$ which was closed under $g$, contrary to the hypothesis. Hence the only elements in $N$ are the elements of the $g$-sequence started by $b$. Having shown this, recall that $g$ is a progressive function. Now consider any $x, y$ in $N$. If $x = y$, then $y \subseteq x$. Now suppose $x \neq y$. Let $x$ be the $m$th term of the $g$-sequence and $y$ the $n$th term. Thus $m \neq n$. On the one hand, suppose that $m < n$. An obvious induction argument

shows that $x \subseteq y$. Now either $g(x) = y$ or not. In either case, $g(x) \subseteq y$ is obvious. On the other hand, suppose that $n < m$. Then the same argument shows that $y \subseteq x$, thus completing the proof of (1).

Here is the "obvious inductive argument" mentioned above (I will refer to this as Lemma 1). We prove that, for all $n$, all $m$th terms where $m < n$ are subsets of the $n$th term of the $g$-sequence. The base case is obvious (as there are no terms of index less than 1). Now suppose the result holds for the $n$th term $x$. Then all terms of index $m$ where $m < n$ are subsets of $x$. But then, because $g$ is progressing, $x$ is a subset of $g(x)$ (the $n+1$th term), and because all the terms of index $m$ where $m < n$ are subsets of $x$, it follows that all terms of index $m$ where $m < n+1$ are subsets of $g(x)$, completing the proof.

We now prove (2) of the problem. We prove by induction on $n$ that every non-empty subset of $N$ that is bounded by the $n$th term of $N$ (that is, the $n$th term of the $g$-sequence $b, g(b), g(g(b)), ...$) has a greatest element. The base case is when $n = 0$, that is, when the term is $b$. Now consider some bounded subset $c$ of $N$ which is bounded by $b$. Thus $c \subseteq N$ and, for all $d \in c$, $d \subseteq b$. But, by Lemma 1, $b$ is a subset of all elements of $N$. Thus $b$ is a subset of all elements $d \in c$. But $d \subseteq b$. Hence $b = d$ for all elements $d \in c$, meaning $c$ is just the set $\{b\}$. Then, of course, $b$ is the greatest element of $c$, therefore proving the base case. Now suppose the hypothesis holds for some $n$. That is, every non-empty subset of $N$ bounded by the $n$th term $x$ of $N$ has a greatest element. Now, the $n+1$th term of $N$ is just $g(x)$. Consider any non-empty subset $B$ of $N$ bounded by $g(x)$. If $g(x) \in B$, then of course $g(x)$ is the greatest element of $B$. So suppose that $g(x) \notin B$. Then every element of $B$ is a proper subset of $g(x)$. Consider any element $y$ of $B$. Then $y \subset g(x)$, meaning $g(x) \subseteq y$ doesn't hold, and therefore $y \subseteq x$ holds (because either $g(x) \subseteq y$ or $y \subseteq x$ by (1) of this problem). But this means that $B$ is bounded by $x$; hence, by the induction assumption, it has a greatest element.

I was not entirely sure what (3) was trying to prove: I interpreted it to mean that, if $x$ is the greatest element of $N$, then it is a fixed point of $g$. Suppose that $x$ is the greatest element of $N$. Then, for all elements $y$ of $N$, $y \subseteq x$. Now, we know that $g(x) \in N$. Because $x$ is the greatest element of $N$, $g(x) \subseteq x$. But also, because $g$ is progressing, $x \subseteq g(x)$. Thus $g(x) = x$, meaning $x$ is a fixed point of $g$.

Now we prove (4) of the problem. Because we have shown that $g(x) \subseteq y$ or $y \subseteq x$ for all $x, y$ in $N$, it follows that $x \subseteq y$ or $y \subseteq x$ (as $x \subseteq g(x)$). Thus $N$ is a nest. It remains to be shown that every non-empty subclass of $N$ has a least element. So consider some non-empty subclass $A$ of $N$. Because $N$ is just the set of all terms of the $g$-sequence $b, g(b), g(g(b)), ...$, it follows that $A$ contains only members of that sequence. Hence $A$ contains some term $x$ whose index (in the $g$-sequence) $n$ is less than or equal to the indices of all the other elements of $A$. By Lemma 1, it follows that $x$ is a subset of all terms with an index greater than or equal to $n$, and hence $x$ is a subset of all other elements of $A$, meaning it is the least element of $A$.

Because $b$ starts the $g$-sequence $b, g(b), g(g(b)), ...$, which makes up the entirety of $N$, Lemma 1 above shows that $b$ is the least element of $N$ (is a subset of all elements of $N$). This proves (5) of the problem.

The reason that this generalizes some of the earlier results of the section is that it takes away the requirenment that $N$ be minimally inductive under $g$. For example, Theorem 4.7 holds now when $N$ is closed under $g$, where $g$ is progressing, and that $b$ is an element of $N$ and that no propert subclass of $N$ containing $b$ is closed. That is, we do not need to know that $N$ is minimally inductive under $g$.

## Exericse 4.2

(I did not think that this proof requires $A$ to be a set)
Suppose that $A$ is inductive under $g$ and that $A$ is a set. Thus we know
(1) $\emptyset \in A$,
(2) for every $x$ in $A$, the set $g(x)$ is in $A$.
Consider the $g$-sequence $\emptyset, g(\emptyset), g(g(\emptyset)), ....$ Because $A$ is inductive under $g$, that $g$-sequence is a subset of

$A$. But clearly, that $g$-sequence is minimally inductive under $g$. Hence there is some subset of $A$ that is minimally inductive under $g$.

While we cannot show that every $g$-inductive class is a set (as Smullyan remarked, our axioms are not yet powerful enough to show that every denumerable class is a set), we can show that every element of every $g$-inductive class is a set. Recall that the definition of a function relied upon the definition of relations, which in turn relied upon the notion of an ordered pair $\langle x, y\rangle$, which was only defined when $x$ and $y$ are sets. Thus $x$ can only be in the domain of a function if $x$ is a set. Now consider any $g$-inductive class $A$, and consider any element $x$ in $A$. Then we know that $g(x) \in A$, and thus, because $x$ is in the domain of $g$, $x$ is a set. If it happens that $A$ is finite (which could occur if, for instance, $\emptyset$ was a fixed point of $g$ and $A$ only contained $\emptyset$), $A$ would be a set as a consequence of Exercise 6.1. Indeed, there are non-denumerable $g$-inductive classes, for instance, the class of all sets, which is $g$-inductive under the power function.

# Section 5

## Exercise 5.1

Suppose that the natural number $n$ is in $A$ and that for any number $x$ in $A$, $x^+$ is also in $A$. We can prove that $A$ contains every number $\geq n$ by $g$-induction. In particular, consider the property $P$: If $m \geq n$, then $m \in A$. We show that this prpoerty holds for all numbers $m$. Either $\emptyset = n$ or $\emptyset < n$. In the first case, $P$ holds, while in the second case, $P$ vacuously holds. Thus $P$ holds for $\emptyset$. Now suppose that $P$ holds for $x$. There are two cases: either $x^+ < n$, $x^+ = n$, or $x^+ > n$. In the first case, $P$ vacuously holds. In the second case, because we assumed $n \in A$, $P$ holds. In the last case, by the induction assumption that $x \in A$, $x^+ \in A$, and thus $P$ holds for $x^+$. Thus, in any case, $P$ holds for $x^+$, therefore completing the induction.

## Exercise 5.2

We suppose $f$ is a function from $\omega$ into $A$.

(a): We suppose that, for all $n$, $f(n) \subseteq f(n^+)$. We can prove that, for all $n$ and $m$, if $n \leq m$ then $f(n) \subseteq f(m)$ by induction on $m$. In particular, we show that, for all numbers $m$, for all numbers $n \leq m$, $f(n) \subseteq f(m)$. The base case is when $m = \emptyset$. The only natural number $\leq \emptyset$ is $\emptyset$, which is a subset of itself, meaning the base case holds. Now suppose that the hypothesis holds for some $m$. Thus, for all $n \leq m$, $f(n) \subseteq f(m)$. Now, we are given that $f(m) \subseteq f(m^+)$. But this means that, for all number $n \leq m$, $f(n) \subseteq f(m^+)$. Thus all numbers $n \leq m^+$ are such that $f(n) \subseteq f(m^+)$, therefore completing the induction.

(b): We prove that, for all $n$, if $f(n) \subset f(n^+)$, then for all $n$ and $m$, if $n < m$, then $f(n) < f(m)$. The base case is when $m = \emptyset$. In this case, there are no natural numbers less than $\emptyset$, so the property vacuously holds. No suppose it holds for some $m$. Thus, for all $n < m$, $f(n) \subset f(m)$. Now, we are given that $f(m) < f(m^+)$. But this means that, for all $n < m$, $f(n) \subset f(m^+)$, thus completing the induction.

# Section 6

## Exercise 6.1

(1): We prove that every finite class is a set. This can be done using induction on the number of elements $n$ in the class. The base case is for the class $\emptyset$, which is a set by $A_3$. Now suppose the result holds for some $n$. Consider some class $S$ of size $n+1$. Pick some arbitrary element $a$ of $S$. By the induction assumption, $S - \{a\}$ is a set. Also, $S = \{a\} \cup (S - \{a\})$. But $\{a\} \cup (S - \{a\}) = \cup\{\{a\}, (S - \{a\})\}$, which is a set by $A_5$ (because $\{\{a\}, (S - \{a\})\}$ is a set by $A_4$), thus proving $S$ is a set.

(2): We prove that every subset of a finite set is finite. This can be done using induction on the number of elements $n$ in the set. The base case is for $\emptyset$, whose only subset is $\emptyset$, which is finite. Thus the base case holds. Now assume the result holds for some $n$. Consider some finite set of $S$ size $n+1$. Consider any subset $F$ of $S$. Either it is a subset of a set of size $n$, or it is not. If it is, the induction assumption shows that it is finite. Otherwise, the only way that it could not be a subset of a set of size $n$ would be if $F$ was $S$ itself (because $S$ only has $n+1$ elements), meaning $F$ would be finite, therefore completing the proof.

(3): We prove that every non-empty finite set of natural numbers has a greatest element. This can be done using induction on the natural numbers. The base case is for $\{\emptyset\}$, whose only element, and therefore greatest element, is $\emptyset$. Now assume that the result holds for some $n$, that is, $n$ has a greatest element. Now consider $n+1$. It is just $n \cup \{n\}$. Thus its elements are either elements of $n$, or $n$ itself. But we know that all elements of $n$ are subsets of $n$ (by results in section 5). Thus all elements of $n+1$ are subsets of $n$, therefore completing the proof.

## Exercise 6.2

We prove that the union of any two finite sets is finite. An equivalent thing to prove is that the statement that the union of a set $S$ with $n$ elements and any other finite set is finite, holds for all $n$. This can be proven using complete induction on $n$. The base case, when $n = 0$, is when $S = \emptyset$. Now consider any other finite set $A$ with $m$ elements. But, because $\emptyset \cup A = A$ holds for all sets $A$, and because $A$ was assumed to be finite, $\emptyset \cup A$ is also finite. Thus the base case is proven. Now suppose the result holds for all numbers less than $n$, that is, for any set $S$ having less than $n$ elements, $S \bigcup A$ has a finite amount of elements for all finite sets $A$. Now consider some set $B$ having $n$ elements. Pick some arbitrary element $a$ of $B$. Then $B = (B - \{a\}) \cup \{a\}$. Now consider any finite set $C$. Then $C \cup B = C \cup ((B - \{a\}) \cup \{a\})$. But, in general, for any sets $X, Y$ and $Z$, $X \cup (Y \cup Z) = (X \cup Y) \cup Z$. Thus $C \cup B = (C \cup (B - \{a\})) \cup \{a\}$. But $B - \{a\}$ has $n-1$ elements. Thus, by the induction assumption, $C \cup (B - \{a\})$ is finite. Also, $\{a\}$ is finite. Thus we know by the induction assumption that, because $C \cup (B - \{a\})$ has been shown to be finite, $(C \cup (B - \{a\})) \cup \{a\}$ is finite, that is, $C \cup B$ is finite, which completes the induction.

## Exercise 6.3

We prove that the union of any finite set $S$ of finite sets is finite. We prove this by induction on the size $n$ of $S$. In the base case, $S$ has only one element $a$, and its union is just all of the elements of $a$, which is finite, as $a$ was assumed to be finite. Now suppose the result holds for some $n$. Consider some set $S$ of size $n+1$ of finite sets. Let its elements be $a_1, a_2, ..., a_n, a_{n+1}$. Let $S_1 = \{a_1, a_2, ..., a_n\}$. Now, $\cup S = (\cup S_1) \cup a_{n+1}$ (as can easily be shown). By the induction assumption, $\cup S_1$ is finite. Then, by Exercise 6.2, $(\cup S_1) \cup a_{n+1}$ is also finite. Hence $\cup S$ is finite, therefore completing the proof.

### Exercise 6.4

Suppose that $M$ is minimally under $g$ where $g$ is progressing. We prove that for every $x$ in $M$, the set of all $y \in M$ such that $y \subseteq x$ is finite. This can be done by using $g$-induction on $M$. For $\emptyset$, it only has a subset of $\emptyset$, meaning the set of all $y \in M$ such that $y \subseteq x$ is finite. Now suppose the result holds for some $x$. That is, the set $S$ of all $y \in M$ such that $y \subseteq x$ is finite. Now consider $g(x)$. Because $g$ is progressing, $x \subseteq g(x)$. Thus all elements of $S$ are subsets of $g(x)$. Now suppose there was some $y \in M$ such that $y \subseteq g(x)$, but $y$ is not a member of $S$ (that is, $y$ is not a subset of $x$). Thus, because $M$ is a nest (as was shown in section 4), $x \subseteq y$. Now, we know that $y \subseteq g(x)$. Thus $x \subseteq y \subseteq g(x)$, meaning, by Theorem 4.10 part (1), either $y = x$ or $y = g(x)$. Thus the elements of the set of all $y \in M$ such that $y \subseteq g(x)$ are the elements of $S$, and the sets $x$ and $g(x)$. But $x \in S$. Thus the set of all $y \in M$ such that $y \subseteq g(x)$ is just the set $S$ together with $g(x)$, meaning it is also a finite set (because $S$ is finite).

We can also show the result using $g$-sequences: Suppose that $M$ is minimally under $g$ where $g$ is progressing. We prove that for every $x$ in $M$, the set of all $y \in M$ such that $y \subseteq x$ is finite. Now, $M$ is just the set of all members of the $g$-sequence $\emptyset, g(\emptyset), g(g(\emptyset)), ....$ We prove that for all $n$, the $n$th term in the above $g$-sequence satisfies the desired conditions. This shows that all elements of $M$ satisfy the desired conditions. Consider the $n$th term in the sequence $x$. Because $g$ is progressing, we know that every $m$th term of the sequence where $m \leq n$ is a subset of the $n$th term. These elements of course form a finite set, one with $n-1$ elements (call this set $S$). Now, there are no other elements of $M$ which are subsets of $x$. For suppose there was one, call it $y$. Then it would have an index greater than $n$ (because all the terms with an index $\leq n$ were already shown to be subsets of $x$). But then, because $g$ is progressing, $x \subseteq y$. But also $y \subseteq x$, meaning $x = y$, proving that $y$ is in $S$. Thus the set of all $y \in M$ such that $y \subseteq x$ is just $S$, which is finite, therefore completing the proof.

### Exercise 6.5

Suppose $M$ is minimally inductive under $g$, where $g$ is progressing. Suppose also that $M$ contains an element $x$ such that $x = g(x)$. By Theorem 4.14, we then know that $x$ is the greatest element of $M$, that is, for all $y \in M$, $y \subseteq M$. But by Exercise 6.4, the set of all $y \in M$ such that $y \subseteq x$ is finite. Then, because every element of $M$ is a subset of $x$, it follows that $M$ is finite.

We can also prove this using $g$-sequences: Suppose $M$ is minimally inductive under $g$, where $g$ is progressing. Suppose also that $M$ contains an element $x$ such that $x = g(x)$. We show that $M$ is finite. We know that $M$ is just the set of all the members in the $g$-sequence $\emptyset, g(\emptyset), g(g(\emptyset)), ....$ Suppose the first element $x$ of $M$ for which $x = g(x)$ has an index $n$ in that $g$-sequence. Then, as can easily be shown, for all terms $y$ with an index greater than $n$ in the sequence, $y = x$. Thus the only elements of $M$ are the first $n$ terms of the $g$-sequence, meaning $M$ is finite, in particular, a set with $n$ elements.

## Section 7

### Exercise 7.1

We can prove the equivalent statement that, if a set $S$ of natural numbers is not finite, then it is denumerable. Note that, because $\omega$ is the set of all natural numbers, $S \subseteq \omega$. We can make a 1-1 correspondence between $S$ and $\omega$ as follows: pick some element $a_0$ of $S$ and pair it with 0. Then pick another element $a_1$ of $S$ and pair it with 1, and, in general, we pair $n$ with some element $a_n$ of $S$, where $a_n$ is disitinct from all $a_m$ where $m < n$. If there was any $n$ for which it could not be paired to an $a_n$, this would

mean that $S$ only had $n-1$ elements, contrary to the assumption that it was infinite. If there was ever some $a_n$ for which no $n$ could be found, this would contradict the fact that $S \subseteq \omega$. Thus this really is a 1-1 correspondence.

We can also show that every non-denumerable set has a denumerable proper subset. For consider any non-denumerable set $A$. We can show that some proper subset of $A$ can be put into a 1-1 correspondence with $\omega$ in the following way: Pick some element $a_0$ of $A$, and pair it with 0. Then pick another element $a_1$ of $A$, and pair it with 1, and in general, for every $n$, we pick a new element $a_n$ of $n$ to match with $n$. If for any $n$ we could not pick a new element of $A$ to match with it, that would mean that $A$ would only have $n-1$ elements, contrary to the assumption that it was infinite. Thus, for every $n$, we can pick an element of $A$ to match with it, meaning we have a denumerable subset $S$ of $A$ consisting of the elements $a_0, a_1, ..., a_n, ....$ If this was the entirety of $A$, then $A$ itself would be denumerable, contrary to the assumption that it was not. Thus $S$ is a proper subset of $A$, therefore completing the proof.

Note the following incorrect proof: Having shown the above result, we can now prove that any infinite subset of a denumerable set is denumerable. For consider some denumerable set $A$. Suppose it had a non-denumerable subset $B$. By the above result, there is a proper subset $C$ of $B$ which is denumerable. Because $B \subseteq A$ and $C \subset B$, $C \subset A$. Thus we would have $A$ being in a 1-1 correspondence with $C$, a proper subset of $A$, which is impossible. Hence $A$ can have no non-denumerable subset. This is of course incorrect because it is not true that $A$ cannot be in a 1-1 correspondence with a proper subset of itself. For instance, $\omega$ can be placed in a 1-1 correspondence with the set of all even numbers.

Note the following incorrect proof that any denumerable set can be put into a 1-1 correspondence with any non-denumerable set: Consider any denumerable set $A$ and any non-denumerable set $B$. Arrange the elements of $A$ in the order $a_0, a_1, ..., a_n, ....$ If for any $n$, $a_n$ did not have a matching element in $B$, then $B$ would be finite, a contradiction. Also, if we ever encountered an element in $B$ for which we could pair no element of $A$ with, this would mean $A$ would be finite, a contradiction. The error in this proof is that we of course can reach an element in $B$ for which we could pair no element of $A$ with (and this does not mean that $A$ is finite; it would only mean so if the statement was:"if we ever encountered some *nth element* in $B$ for which we could pair no element of $A$ with" This would then mean that $A$ was finite). Of course, this element in $B$ that we could not pair up with $A$ would come after pairing off a denumerable amount of elements from $B$, meaning we would already know that $A$ was infinite (and thus there is no supposed contradiction).

We can prove the fact that no denumerable subset can have a non-denumerable subset in a different way. First, define a set $S$ to be *greater* than a set $F$ if $S$ cannot be put into a 1-1 correspondence with $F$, but some proper subset of $S$ can be put into a 1-1 correspondence with $F$. If $S$ is greater than $F$, we denote this by $S > F$, and say that $F$ is less than $S$, and denote this by $F < S$. We first show some facts about sets being greater than one another, and then the desired result comes easily.

First we show that, for any sets $S$ and $F$, either $S < F$ or $F < S$, or $S$ and $F$ can be put into a 1-1 correspondence. Suppose that $S$ and $F$ cannot be put into a 1-1 correspondence. Pick some element $a_1$ of $S$, and $b_1$ of $F$, and pair them together. Then pick some new element $a_2$ of $S$, and some new element $b_2$ of $F$, and so on. Becaus $S$ and $F$ cannot be put into a 1-1 correspondence, we will at some point either reach some element of $F$ for which no element of $S$ can be found, in which case $F > S$ (as the the elements paired up with $S$ up until that point represent a subset of $F$, and is in a 1-1 correspondence with $S$), or we reach some element of $S$ for which no element of $F$ can be found, in which case $S > F$.

**Lemma**: *For all sets $S$, no subset of $S$ can be greater than $S$.*
For suppose some subset $F$ was greater than $S$. Then $S$ and $F$ could not be put into a 1-1 correspondence, but some proper subset of $F$ could be put into a 1-1 correspondence with $S$. We can show that $F$ and $S$ can, in fact, be put into a 1-1 correspondence, in the following way: If there was ever some element of $F$ for which we could not find any element of $S$ to pair it with, this would contradict the fact that $F$ was a subset of $S$. If there was ever some element of $S$ for which we could not find any element of $F$ to pair it with, this contradicts the fact that some proper subset of $F$ could be put into a 1-1 correspondence with $S$. Thus $S$

and $F$ can be put into a 1-1 correspondence, a contradiction, therefore completing the proof.

After this, we can prove the desired result about denumerable sets (that no denumerable set can have a non-denumerable subset). For suppose we had some denuemrable set $A$ having a non-denumerable subset $B$. We have already shown that some proper subset of every non-denumerable subset is denumerable. Hence a proper subset of $B$ is denumerable (can be put into a 1-1 correspondence with $A$). But, by definition, $B$ cannot be put into a 1-1 correspondence with $A$. Thus $B > A$, which contradicts the Lemma, therefore completing the proof.

We now show that, for any sets $S$ and $F$, it is not the case that both $S > F$ and $F > S$ hold. For suppose the hypothesis did not hold. Then we would have $S$ and $F$ not in a 1-1 correspondence, and some proper subset $A$ of $S$ being in a 1-1 correspondence with $F$, and some proper subset $B$ of $F$ being in a 1-1 correspondence with $S$. Then we would have $A$, a proper subset of $S$, being put into a 1-1 correspondence with $F$, a set which is greater than $S$. Thus $A$, a subset of $S$, is greater than $S$, which is impossible by the Lemma, therefore completing the proof.

The following Lemma will be useful: **Union Lemma**: *For any sets $A, B, C$, and $D$, if $A$ and $D$ can be placed in a 1-1 correspondence, and $B$ and $C$ can be placed in a 1-1 correspondence, and $A$ and $B$ contain no elements in common, and $C$ and $D$ contain no elements in common, then $A \cup B$ can be placed in a 1-1 correspondence with $C \cup D$.*

**Proof**: Suppose we have sets $A, B, C$, and $D$ in the relation described in the statement of the Lemma. First, we pair off the elements of $A$ in the set $A \cup B$ with the elements of $C$ in the set $C \cup D$, which can be done, as $A$ and $C$ can be placed in a 1-1 correspondence. Also, this does not use any elements of $B$, as $A$ and $B$ were assumed to be disjoint. Then we can do the same thing for $B$ and $D$, therefor showing we can pair up all elements of $A \cup B$ with $C \cup D$, meaning $A \cup B$ can be placed in a 1-1 correspondence with $C \cup D$.

We now show that any infinite set can be put into a 1-1 correspondence with a proper subset of itself. We first prove this for denumerable sets, and then the case for non-denumerable sets easily follows. Consider any denumerable set $S$. Thus the elements of $S$ can be arranged in a denumerable sequence $a_1, a_2, ..., a_n, ....$ Now consider the set $S - \{a_1\}$, which is a proper subset of $S$. We can show that it can be placed in a 1-1 correspondence with $S$ as follows: Note that the elements of $S - \{a_1\}$ are $b_1, b_2, ..., b_{n+1}, ...$, where $b_1 = a_2$, $b_2 = a_3$, and, in general, $b_n = a_{n+1}$. We pair up $a_1$ with $b_1$, $a_2$ with $b_2$, and, in general, $a_n$ with $b_n$, thus showing that there is a 1-1 correspondence between $S$ and $S - \{a_1\}$, a proper subset of $S$.

Now that we have shown the denumerable case, consider some non-denumerable set $S$. We have already shown that $S$ contains some denumerable subset $D$. Now pick some arbitrary element $a$ of $D$. Now, $D \cup (S - D) = S$. Consider the set $(D - \{a\}) \cup (S - D)$ (which is just the set $S - \{a\}$), which is a proper subset of $S$. We have already shown that $D - \{a\}$ can be placed into a 1-1 correspondence with $D$. Then, because $D - \{a\}$ can be placed in a 1-1 correspondence with $D$, $S - D$ can be placed in a 1-1 correspondence with $S - D$, $D$ and $S - D$ are disjoint, and $D - \{a\}$ and $S - D$ are disjoint, the Union Lemma shows that $D \cup (S - D)$ can be placed in a 1-1 correspondence with $(D - \{a\}) \cup (S - D)$, that is, $S$ can be placed in a 1-1 correspondence with a proper subset of itself.

Now we prove that, for any element $a$ of any infinite set $S$, there is a denumerable subset of $S$ containing $a$. We have already shown that $S$ must contain some denumerable set $D$. Either $D$ contains $a$ or it doesn't. If it does, we are done. If it doesn't, pick some arbitrary element $e$ of $D$. Then $(D - \{e\}) \cup \{a\}$ is the desired set, as we can show that it is denumerable. We know that the elements of $D$ can be placed in a denumerable sequence $a_0, a_1, ..., a_n, ....$ If $a_0 \neq e$, then we pair 0 off with $a_0$, and if $a_0 = e$, then we pair 0 off with $a$. If $a_1 \neq e$, then we pair 1 off with $a_1$, and if $a_1 = e$, then we pair 1 off with $a$. And, in general, if $a_n \neq E$, then we pair 1 off with $a_n$, and if $a_n = e$, then we pair $n$ off with $a$, thus showing that $(D - \{e\}) \cup \{a\}$ is denumerable.

Now we prove that the union of any denumerable set and any finite set is denumerable. For consider some finite set $F$ and some denumerable set $D$. Then the elements of $D$ can be arranged in the sequence $a_1, a_2, ..., a_n, ....$ Also, the set $G$ of all elements of $F$ which are not already elements of $D$ is finite, say its elements are in the sequence $b_1, b_2, ..., b_m$. Then $D \cup F = D \cup G = \{b_1, b_2, ..., b_m, a_1, a_2, ..., a_n, ...\}$. We pair off $b_1$ with the $a_1 \in S$, $b_2$ with the $a_2 \in S$, and, for all $i \leq m$, we pair off $b_i$ with the $a_i \in S$. Then, we pair off the $a_1$ in $D \cup G$ with the $a_{m+1}$ in $D$, the $a_2$ in $D \cup G$ with the $a_{m+2}$ in $D$, and, in general, for all $i > m$, we pair off the $a_i \in D \cup G$ with the $a_{m+i} \in D$, thus showing that $D \cup G$ can be placed in a 1-1 correspondence with $D$.

Now we prove that the union of any non-denumerable set $S$ (thus proving the result for all infintie sets) and any finite set can be placed in a 1-1 correspondence with $S$. For consider some non-denumerable set $S$ and some finite set $F$. Let $G$ be the set of all elements in $F$ not in $S$. Now, we know that $S$ contains some denumerable subset $D$. By the preceeding result, $D \cup G$ is denumerable, that is, $D \cup G$ can be placed in a 1-1 correspondence with $D$. Then, because $S - D$ can be placed in a 1-1 correspondence with $S - D$, and $D \cup G$ can be placed in a 1-1 correspondence with $D$, and $D \cup G$ and $S - D$ are disjoint (because $G$ and $S$ are disjoint), and $D$ and $S - D$ are disjoint, the Union Lemma shows that $(D \cup G) \cup (S - D)$, that is, $S \cup G$, can be placed in a 1-1 correspondence with $D \cup (S - D)$, that is, $S$, therefore completing the proof.

# Section 8

## Exercise 8.1

Suppose that $A$ is a non-empty set of natural numbers having no greatest element. We prove that $A$ is denumerable. We have already shown (in Exercise 7.1) that $A$ is either finite or denumerable. It suffices to show that every non-empty finite set of natural numbers has a greatest element, which would show that $A$ must be denumerable. We prove this by induction on the size $n$ of the finite set $S$. The base case is when $n = 1$. In this case, $S = \{m\}$ for some natural number $m$, meaning the greatest element of $S$ is just $m$. Now suppose the result holds for some $n$, that is, any finite set of natural numbers of size $n$ has a greatest element. Now consider any finite set $F$ of natural numbers $S$ having $n + 1$ elements. Pick some arbitrary element $a$ of $F$. Then $S - \{a\}$ has $n$ elements. By the induction assumption, it has a greatest element $g$. Now, either $g \subseteq a$ or $a \subseteq g$. In the first case, because $g$ is the greatest element of $S - \{a\}$, all elements of $S - \{a\}$ are subsets of $a$. But the only elements of $S$ are those of $S - \{a\}$ and $a$. Thus $a$ is the greatest element of $S$. In the second case, because the only elements of $S$ are those of $S - \{a\}$ and $a$, and because all of these are subsets of $g$, $g$ is the greastest element of $S$. Thus, in any case, $S$ has a greatest element, meaning, by induction, all non-empty finite sets of natural numbers have a greatest element. Thus, because $A$ does not have a greatest element, it follows that $A$ is denumerable.